

Supplementary Material: Detailed Benchmark Results

This supplement reports detailed benchmark results for all datasets and both labelling strategies (`mvc`, `classic`). Each cell shows mean $\pm$ standard deviation over 10 seeds. Columns labelled 0.2, 0.5, and 0.8 refer to target labelling rates $c_{\text{calc}} \in \{0.2, 0.5, 0.8\}$.

1 Labelling Strategy: `mvc`

Table 1: Detailed results for `ISAP_corpus` with labelling strategy `mvc` (mean $\pm$ std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.089 $\pm$ 0.017	0.127 $\pm$ 0.018	0.249 $\pm$ 0.025	0.005	0.005	0.005	0.000	0.000	0.000
Clust	0.772 $\pm$ 0.043	0.104 $\pm$ 0.025	0.184 $\pm$ 0.030	0.022 $\pm$ 0.003	0.005	0.005	0.017	0.000	0.000
S-LassClust	0.820 $\pm$ 0.051	0.090 $\pm$ 0.021	0.144 $\pm$ 0.023	0.036$\pm$0.008	0.005	0.005	0.017	0.000	0.000
NS-LassClust	0.821$\pm$0.050	0.090 $\pm$ 0.021	0.144 $\pm$ 0.023	0.036$\pm$0.008	0.005	0.005	0.017	0.000	0.000
LassoJoint	0.144 $\pm$ 0.036	0.176 $\pm$ 0.021	0.241 $\pm$ 0.009	0.007	0.005	0.005	0.004 $\pm$ 0.001	0.000	0.000
Spy	0.580 $\pm$ 0.039	0.543$\pm$0.036	0.543$\pm$0.036	0.010 $\pm$ 0.001	0.009$\pm$0.001	0.009$\pm$0.001	0.019$\pm$0.001	0.019	0.019

Table 2: Detailed results for `job_offers_pl_corpus` with labelling strategy `mvc` (mean $\pm$ std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.575$\pm$0.012	0.667 $\pm$ 0.012	0.798$\pm$0.009	0.178$\pm$0.005	0.221 $\pm$ 0.008	0.355 $\pm$ 0.012	0.000	0.000	0.001 $\pm$ 0.002
Clust	0.345 $\pm$ 0.012	0.587 $\pm$ 0.009	0.663 $\pm$ 0.009	0.120 $\pm$ 0.002	0.195 $\pm$ 0.009	0.256 $\pm$ 0.014	0.287$\pm$0.001	0.040 $\pm$ 0.014	0.090 $\pm$ 0.021
S-LassClust	0.327 $\pm$ 0.006	0.684 $\pm$ 0.012	0.758 $\pm$ 0.013	0.117 $\pm$ 0.001	0.305$\pm$0.020	0.383$\pm$0.024	0.286 $\pm$ 0.001	0.106 $\pm$ 0.021	0.215 $\pm$ 0.019
NS-LassClust	0.325 $\pm$ 0.008	0.684 $\pm$ 0.012	0.758 $\pm$ 0.013	0.117 $\pm$ 0.001	0.305$\pm$0.020	0.383$\pm$0.024	0.286 $\pm$ 0.001	0.106 $\pm$ 0.021	0.215 $\pm$ 0.019
LassoJoint	0.436 $\pm$ 0.011	0.704$\pm$0.019	0.766 $\pm$ 0.013	0.158 $\pm$ 0.001	0.298 $\pm$ 0.020	0.371 $\pm$ 0.019	0.116 $\pm$ 0.016	0.169 $\pm$ 0.023	0.312$\pm$0.017
Spy	0.484 $\pm$ 0.009	0.522 $\pm$ 0.011	0.522 $\pm$ 0.010	0.154 $\pm$ 0.003	0.170 $\pm$ 0.005	0.170 $\pm$ 0.004	0.032 $\pm$ 0.028	0.274$\pm$0.008	0.271 $\pm$ 0.006

Table 3: Detailed results for `open_subtitles_corpus` with labelling strategy `mvc` (mean $\pm$ std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.565$\pm$0.012	0.677 $\pm$ 0.011	0.819 $\pm$ 0.012	0.682$\pm$0.011	0.761 $\pm$ 0.012	0.874 $\pm$ 0.012	0.000	0.105 $\pm$ 0.009	0.819 $\pm$ 0.006
Clust	0.532 $\pm$ 0.010	0.488 $\pm$ 0.011	0.492 $\pm$ 0.010	0.633 $\pm$ 0.007	0.604 $\pm$ 0.008	0.606 $\pm$ 0.007	0.357 $\pm$ 0.013	0.751$\pm$0.006	0.777 $\pm$ 0.003
S-LassClust	0.555 $\pm$ 0.013	0.521 $\pm$ 0.011	0.571 $\pm$ 0.010	0.645 $\pm$ 0.009	0.627 $\pm$ 0.008	0.664 $\pm$ 0.009	0.368 $\pm$ 0.037	0.746 $\pm$ 0.006	0.783 $\pm$ 0.004
NS-LassClust	0.558 $\pm$ 0.013	0.521 $\pm$ 0.011	0.571 $\pm$ 0.010	0.648 $\pm$ 0.009	0.627 $\pm$ 0.008	0.664 $\pm$ 0.009	0.387$\pm$0.022	0.746 $\pm$ 0.006	0.783 $\pm$ 0.004
LassoJoint	0.510 $\pm$ 0.012	0.826$\pm$0.014	0.907$\pm$0.008	0.646 $\pm$ 0.006	0.865$\pm$0.014	0.919$\pm$0.009	0.222 $\pm$ 0.043	0.626 $\pm$ 0.012	0.891$\pm$0.006
Spy	0.488 $\pm$ 0.005	0.492 $\pm$ 0.010	0.490 $\pm$ 0.009	0.636 $\pm$ 0.002	0.617 $\pm$ 0.006	0.617 $\pm$ 0.006	0.005 $\pm$ 0.003	0.019 $\pm$ 0.004	0.018 $\pm$ 0.004

Table 4: Detailed results for `plwiki` with labelling strategy `mvc` (mean $\pm$ std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.665 $\pm$ 0.010	0.643 $\pm$ 0.010	0.721 $\pm$ 0.008	0.525 $\pm$ 0.009	0.517 $\pm$ 0.009	0.626 $\pm$ 0.013	0.005 $\pm$ 0.002	0.023 $\pm$ 0.005	0.029 $\pm$ 0.008
Clust	0.124 $\pm$ 0.005	0.661 $\pm$ 0.012	0.706 $\pm$ 0.011	0.262 $\pm$ 0.001	0.567 $\pm$ 0.011	0.626 $\pm$ 0.011	0.587 $\pm$ 0.001	0.100 $\pm$ 0.012	0.122 $\pm$ 0.014
S-LassClust	0.153 $\pm$ 0.011	0.688 $\pm$ 0.011	0.733 $\pm$ 0.009	0.266 $\pm$ 0.002	0.598 $\pm$ 0.011	0.662 $\pm$ 0.011	0.588 $\pm$ 0.001	0.301 $\pm$ 0.013	0.348 $\pm$ 0.017
NS-LassClust	0.153 $\pm$ 0.011	0.688 $\pm$ 0.011	0.733 $\pm$ 0.009	0.266 $\pm$ 0.002	0.598 $\pm$ 0.011	0.662 $\pm$ 0.011	0.588 $\pm$ 0.001	0.301 $\pm$ 0.013	0.348 $\pm$ 0.017
LassoJoint	0.729$\pm$0.014	0.781$\pm$0.009	0.853$\pm$0.005	0.590$\pm$0.010	0.727$\pm$0.012	0.812$\pm$0.009	0.691$\pm$0.020	0.620$\pm$0.015	0.643$\pm$0.010
Spy	0.409 $\pm$ 0.022	0.482 $\pm$ 0.005	0.481 $\pm$ 0.005	0.372 $\pm$ 0.011	0.408 $\pm$ 0.003	0.407 $\pm$ 0.002	0.526 $\pm$ 0.019	0.579 $\pm$ 0.003	0.578 $\pm$ 0.003

Table 5: Detailed results for `ulotki_medyczne` with labelling strategy `mvc` (mean $\pm$ std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.173 $\pm$ 0.009	0.545 $\pm$ 0.021	0.827 $\pm$ 0.011	0.035	0.065 $\pm$ 0.004	0.243 $\pm$ 0.020	0.000	0.000	0.000
Clust	0.600 $\pm$ 0.012	0.197 $\pm$ 0.007	0.282 $\pm$ 0.011	0.069 $\pm$ 0.002	0.036	0.040 $\pm$ 0.001	0.159 $\pm$ 0.003	0.001 $\pm$ 0.001	0.004 $\pm$ 0.002
S-LassClust	0.677$\pm$0.019	0.262 $\pm$ 0.009	0.331 $\pm$ 0.011	0.089 $\pm$ 0.006	0.039	0.043 $\pm$ 0.001	0.165$\pm$0.003	0.008 $\pm$ 0.002	0.013 $\pm$ 0.004
NS-LassClust	0.676 $\pm$ 0.020	0.262 $\pm$ 0.009	0.331 $\pm$ 0.011	0.089 $\pm$ 0.007	0.039	0.043 $\pm$ 0.001	0.164 $\pm$ 0.003	0.008 $\pm$ 0.002	0.013 $\pm$ 0.004
LassoJoint	0.306 $\pm$ 0.012	0.799 $\pm$ 0.025	0.951$\pm$0.004	0.062	0.193 $\pm$ 0.026	0.522$\pm$0.032	0.001 $\pm$ 0.001	0.000	0.015 $\pm$ 0.007
Spy	0.633 $\pm$ 0.068	0.854$\pm$0.011	0.851 $\pm$ 0.011	0.124$\pm$0.019	0.225$\pm$0.012	0.220 $\pm$ 0.013	0.000	0.159$\pm$0.034	0.149$\pm$0.031

Table 6: Detailed results for `wolne_lektury_corpus` with labelling strategy `mvc` (mean $\pm$ std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.626$\pm$0.012	0.650 $\pm$ 0.016	0.718 $\pm$ 0.012	0.416$\pm$0.009	0.431 $\pm$ 0.011	0.517$\pm$0.014	0.000	0.002 $\pm$ 0.002	0.054 $\pm$ 0.013
Clust	0.261 $\pm$ 0.009	0.561 $\pm$ 0.014	0.624 $\pm$ 0.016	0.261 $\pm$ 0.002	0.376 $\pm$ 0.008	0.421 $\pm$ 0.012	0.569$\pm$0.003	0.048 $\pm$ 0.010	0.081 $\pm$ 0.011
S-LassClust	0.298 $\pm$ 0.024	0.596 $\pm$ 0.012	0.656 $\pm$ 0.012	0.270 $\pm$ 0.006	0.398 $\pm$ 0.008	0.438 $\pm$ 0.010	0.569$\pm$0.003	0.098 $\pm$ 0.010	0.168 $\pm$ 0.018
NS-LassClust	0.298 $\pm$ 0.023	0.596 $\pm$ 0.012	0.656 $\pm$ 0.012	0.270 $\pm$ 0.005	0.398 $\pm$ 0.008	0.438 $\pm$ 0.010	0.569$\pm$0.003	0.098 $\pm$ 0.010	0.168 $\pm$ 0.018
LassoJoint	0.493 $\pm$ 0.014	0.660$\pm$0.012	0.726$\pm$0.011	0.375 $\pm$ 0.006	0.445$\pm$0.010	0.499 $\pm$ 0.013	0.251 $\pm$ 0.032	0.157 $\pm$ 0.020	0.262 $\pm$ 0.017
Spy	0.498 $\pm$ 0.012	0.534 $\pm$ 0.007	0.533 $\pm$ 0.007	0.375 $\pm$ 0.006	0.395 $\pm$ 0.004	0.394 $\pm$ 0.004	0.520 $\pm$ 0.012	0.557$\pm$0.004	0.556$\pm$0.004

2 Labelling Strategy: classic

Table 7: Detailed results for `ISAP_corpus` with labelling strategy `classic` (mean $\pm$ std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.826$\pm$0.055	0.757$\pm$0.062	0.645$\pm$0.058	0.038$\pm$0.010	0.025$\pm$0.008	0.014$\pm$0.003	0.000	0.000	0.000
Clust	0.640 $\pm$ 0.092	0.598 $\pm$ 0.086	0.491 $\pm$ 0.062	0.012 $\pm$ 0.003	0.011 $\pm$ 0.003	0.008 $\pm$ 0.001	0.000	0.000	0.000
S-LassClust	0.682 $\pm$ 0.068	0.538 $\pm$ 0.052	0.416 $\pm$ 0.046	0.013 $\pm$ 0.002	0.009 $\pm$ 0.001	0.007 $\pm$ 0.001	0.000	0.000	0.000
NS-LassClust	0.682 $\pm$ 0.068	0.558 $\pm$ 0.091	0.416 $\pm$ 0.046	0.013 $\pm$ 0.002	0.010 $\pm$ 0.002	0.007 $\pm$ 0.001	0.000	0.000	0.000
LassoJoint	0.666 $\pm$ 0.077	0.663 $\pm$ 0.081	0.577 $\pm$ 0.042	0.017 $\pm$ 0.008	0.015 $\pm$ 0.005	0.010 $\pm$ 0.001	0.000	0.000	0.000
Spy	0.543 $\pm$ 0.035	0.543 $\pm$ 0.036	0.543 $\pm$ 0.036	0.009 $\pm$ 0.001	0.009 $\pm$ 0.001	0.009 $\pm$ 0.001	0.019	0.019	0.019

Table 8: Detailed results for `job_offers_pl_corpus` with labelling strategy `classic` (mean $\pm$ std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.822 $\pm$ 0.011	0.828$\pm$0.011	0.819 $\pm$ 0.012	0.377 $\pm$ 0.016	0.384 $\pm$ 0.018	0.373 $\pm$ 0.017	0.000	0.000	0.008 $\pm$ 0.004
Clust	0.662 $\pm$ 0.013	0.713 $\pm$ 0.013	0.730 $\pm$ 0.013	0.267 $\pm$ 0.013	0.309 $\pm$ 0.017	0.322 $\pm$ 0.017	0.039 $\pm$ 0.013	0.068 $\pm$ 0.017	0.117 $\pm$ 0.020
S-LassClust	0.732 $\pm$ 0.015	0.762 $\pm$ 0.037	0.808 $\pm$ 0.013	0.343 $\pm$ 0.016	0.379 $\pm$ 0.052	0.438 $\pm$ 0.023	0.061 $\pm$ 0.015	0.169 $\pm$ 0.099	0.225 $\pm$ 0.017
NS-LassClust	0.732 $\pm$ 0.015	0.781 $\pm$ 0.029	0.808 $\pm$ 0.013	0.343 $\pm$ 0.016	0.400$\pm$0.040	0.438 $\pm$ 0.023	0.061 $\pm$ 0.015	0.088 $\pm$ 0.040	0.225 $\pm$ 0.017
LassoJoint	0.892$\pm$0.008	0.803 $\pm$ 0.026	0.888$\pm$0.009	0.635$\pm$0.021	0.398 $\pm$ 0.028	0.632$\pm$0.020	0.590$\pm$0.013	0.233 $\pm$ 0.201	0.577$\pm$0.017
Spy	0.521 $\pm$ 0.010	0.521 $\pm$ 0.010	0.521 $\pm$ 0.010	0.170 $\pm$ 0.004	0.169 $\pm$ 0.004	0.169 $\pm$ 0.004	0.271 $\pm$ 0.006	0.268$\pm$0.005	0.267 $\pm$ 0.006

Table 9: Detailed results for `open_subtitles_corpus` with labelling strategy `classic` (mean $\pm$ std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.704$\pm$0.023	0.852$\pm$0.016	0.856$\pm$0.010	0.786$\pm$0.019	0.893$\pm$0.012	0.892$\pm$0.009	0.004 $\pm$ 0.002	0.131 $\pm$ 0.028	0.853$\pm$0.005
Clust	0.482 $\pm$ 0.011	0.486 $\pm$ 0.010	0.490 $\pm$ 0.010	0.600 $\pm$ 0.007	0.603 $\pm$ 0.007	0.606 $\pm$ 0.007	0.741 $\pm$ 0.006	0.772 $\pm$ 0.003	0.780 $\pm$ 0.002
S-LassClust	0.502 $\pm$ 0.012	0.518 $\pm$ 0.026	0.582 $\pm$ 0.009	0.612 $\pm$ 0.009	0.623 $\pm$ 0.018	0.672 $\pm$ 0.010	0.739 $\pm$ 0.005	0.776$\pm$0.005	0.785 $\pm$ 0.002
NS-LassClust	0.502 $\pm$ 0.012	0.531 $\pm$ 0.013	0.582 $\pm$ 0.009	0.612 $\pm$ 0.009	0.633 $\pm$ 0.011	0.672 $\pm$ 0.010	0.739 $\pm$ 0.005	0.713 $\pm$ 0.163	0.785 $\pm$ 0.002
LassoJoint	0.469 $\pm$ 0.014	0.846 $\pm$ 0.119	0.508 $\pm$ 0.015	0.604 $\pm$ 0.016	0.879 $\pm$ 0.082	0.641 $\pm$ 0.009	0.788$\pm$0.002	0.754 $\pm$ 0.192	0.789 $\pm$ 0.001
Spy	0.490 $\pm$ 0.010	0.489 $\pm$ 0.009	0.488 $\pm$ 0.009	0.617 $\pm$ 0.006	0.617 $\pm$ 0.006	0.617 $\pm$ 0.006	0.018 $\pm$ 0.004	0.018 $\pm$ 0.004	0.018 $\pm$ 0.004

Table 10: Detailed results for `plwiki` with labelling strategy `classic` (mean $\pm$ std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.722 $\pm$ 0.023	0.730 $\pm$ 0.012	0.731 $\pm$ 0.009	0.587 $\pm$ 0.035	0.607 $\pm$ 0.020	0.622 $\pm$ 0.016	0.000	0.000 $\pm$ 0.001	0.004 $\pm$ 0.004
Clust	0.710 $\pm$ 0.023	0.705 $\pm$ 0.016	0.720 $\pm$ 0.011	0.654 $\pm$ 0.027	0.638 $\pm$ 0.020	0.648 $\pm$ 0.015	0.011 $\pm$ 0.005	0.019 $\pm$ 0.007	0.045 $\pm$ 0.011
S-LassClust	0.429 $\pm$ 0.034	0.674 $\pm$ 0.050	0.718 $\pm$ 0.011	0.416 $\pm$ 0.027	0.615 $\pm$ 0.055	0.652 $\pm$ 0.015	0.018 $\pm$ 0.007	0.046 $\pm$ 0.023	0.117 $\pm$ 0.015
NS-LassClust	0.429 $\pm$ 0.034	0.643 $\pm$ 0.044	0.718 $\pm$ 0.011	0.416 $\pm$ 0.027	0.577 $\pm$ 0.048	0.652 $\pm$ 0.015	0.018 $\pm$ 0.007	0.030 $\pm$ 0.013	0.117 $\pm$ 0.015
LassoJoint	0.869$\pm$0.010	0.790$\pm$0.065	0.881$\pm$0.006	0.841$\pm$0.013	0.718$\pm$0.105	0.851$\pm$0.010	0.527 $\pm$ 0.054	0.192 $\pm$ 0.165	0.754$\pm$0.011
Spy	0.481 $\pm$ 0.005	0.480 $\pm$ 0.005	0.480 $\pm$ 0.005	0.407 $\pm$ 0.002	0.407 $\pm$ 0.002	0.407 $\pm$ 0.002	0.578$\pm$0.003	0.578$\pm$0.003	0.578 $\pm$ 0.003

Table 11: Detailed results for `ulotki_medyczne` with labelling strategy `classic` (mean $\pm$ std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.920 $\pm$ 0.004	0.917 $\pm$ 0.005	0.915 $\pm$ 0.006	0.421 $\pm$ 0.022	0.413 $\pm$ 0.023	0.406 $\pm$ 0.024	0.000	0.000	0.005 $\pm$ 0.008
Clust	0.381 $\pm$ 0.012	0.427 $\pm$ 0.012	0.457 $\pm$ 0.010	0.046 $\pm$ 0.001	0.050 $\pm$ 0.001	0.052 $\pm$ 0.001	0.012 $\pm$ 0.002	0.016 $\pm$ 0.003	0.022 $\pm$ 0.005
S-LassClust	0.369 $\pm$ 0.013	0.424 $\pm$ 0.020	0.462 $\pm$ 0.010	0.045 $\pm$ 0.001	0.049 $\pm$ 0.002	0.053 $\pm$ 0.001	0.018 $\pm$ 0.004	0.040 $\pm$ 0.017	0.042 $\pm$ 0.006
NS-LassClust	0.369 $\pm$ 0.013	0.438 $\pm$ 0.017	0.462 $\pm$ 0.010	0.045 $\pm$ 0.001	0.051 $\pm$ 0.002	0.053 $\pm$ 0.001	0.018 $\pm$ 0.004	0.022 $\pm$ 0.009	0.042 $\pm$ 0.006
LassoJoint	0.989$\pm$0.002	0.946$\pm$0.049	0.990$\pm$0.002	0.782$\pm$0.041	0.578$\pm$0.185	0.811$\pm$0.040	0.782$\pm$0.021	0.420$\pm$0.358	0.793$\pm$0.020
Spy	0.849 $\pm$ 0.011	0.847 $\pm$ 0.011	0.845 $\pm$ 0.011	0.217 $\pm$ 0.011	0.214 $\pm$ 0.011	0.211 $\pm$ 0.010	0.148 $\pm$ 0.030	0.145 $\pm$ 0.032	0.141 $\pm$ 0.033

Table 12: Detailed results for `wolne_lektury_corpus` with labelling strategy `classic` (mean $\pm$ std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.693 $\pm$ 0.015	0.729$\pm$0.013	0.748 $\pm$ 0.012	0.478 $\pm$ 0.015	0.523$\pm$0.017	0.555$\pm$0.018	0.000	0.003 $\pm$ 0.002	0.327 $\pm$ 0.030
Clust	0.589 $\pm$ 0.023	0.643 $\pm$ 0.017	0.666 $\pm$ 0.016	0.406 $\pm$ 0.017	0.437 $\pm$ 0.015	0.456 $\pm$ 0.015	0.036 $\pm$ 0.009	0.058 $\pm$ 0.013	0.172 $\pm$ 0.026
S-LassClust	0.438 $\pm$ 0.033	0.621 $\pm$ 0.019	0.664 $\pm$ 0.013	0.320 $\pm$ 0.014	0.425 $\pm$ 0.015	0.458 $\pm$ 0.012	0.046 $\pm$ 0.011	0.090 $\pm$ 0.027	0.279 $\pm$ 0.028
NS-LassClust	0.438 $\pm$ 0.033	0.621 $\pm$ 0.014	0.664 $\pm$ 0.013	0.320 $\pm$ 0.014	0.424 $\pm$ 0.012	0.458 $\pm$ 0.012	0.046 $\pm$ 0.011	0.078 $\pm$ 0.020	0.279 $\pm$ 0.028
LassoJoint	0.767$\pm$0.009	0.709 $\pm$ 0.029	0.776$\pm$0.011	0.544$\pm$0.013	0.488 $\pm$ 0.027	0.554 $\pm$ 0.015	0.601$\pm$0.028	0.299 $\pm$ 0.251	0.681$\pm$0.009
Spy	0.533 $\pm$ 0.007	0.534 $\pm$ 0.007	0.533 $\pm$ 0.007	0.394 $\pm$ 0.004	0.394 $\pm$ 0.004	0.394 $\pm$ 0.004	0.556 $\pm$ 0.004	0.556$\pm$0.004	0.556 $\pm$ 0.004

3 Feature Descriptions

Table 13 provides a comprehensive list of all 22 linguistic and stylistic features used in the benchmark. All numeric features are rescaled to the $[0, 1]$ range using min-max normalization before model training.

Table 13: Description of 22 input features (X) used in the benchmark.

Feature	Description
<code>adj_ratio</code>	Ratio of adjectives to total words
<code>adjectives</code>	Count of adjectives in the document
<code>adverbs</code>	Count of adverbs in the document
<code>avg_sentence_length</code>	Average number of words per sentence
<code>avg_word_length</code>	Average number of characters per word
<code>camel_case</code>	Count of camelCase words
<code>capitalized_words</code>	Count of words starting with uppercase letter
<code>characters</code>	Total character count (excluding whitespace)
<code>gunning_fog</code>	Gunning Fog index; estimates U.S. school grade needed to understand text
<code>lexical_density</code>	Ratio of content words (nouns, verbs, adjectives, adverbs) to total words
<code>noun_ratio</code>	Ratio of nouns to total words
<code>nouns</code>	Count of nouns in the document
<code>oovs</code>	Count of out-of-vocabulary words (not in standard Polish dictionary)
<code>pos_num</code>	Count of numerals (number-tagged part-of-speech tokens)
<code>pos_x</code>	Count of unknown or rare part-of-speech categories
<code>punctuations</code>	Count of punctuation marks
<code>sentences</code>	Count of sentences (detected by sentence boundary markers)
<code>stopwords</code>	Count of common Polish stopwords (high-frequency, low-content words)
<code>symbols</code>	Count of non-alphanumeric symbols (special characters)
<code>verb_ratio</code>	Ratio of verbs to total words
<code>verbs</code>	Count of verbs in the document
<code>words</code>	Total word count